

Tools of the Trade: Linux and SQL

Activity: Working with the Bash Shell

Introduction

In this activity, I practiced some basic commands in the Linux Bash shell. The main purpose of the activity was to understand how we can communicate with the operating system by using commands through the terminal.

The commands I practiced were:

- `echo`
- `expr`
- `clear`

I also used some simple security-related examples with calculations.

Activity Scenario

As a security professional, it is important to know how to interact with an operating system through the command line.

In this activity, I worked with a Linux Bash shell where the `analyst` user was already logged in.

The activity focused on using basic commands and understanding the output returned by the Bash shell.

Lab Environment

- **Course:** Google Cybersecurity Certificate
 - **Course:** Tools of the Trade: Linux and SQL
 - **Activity:** Working with the Bash Shell
 - **Operating System:** Linux
 - **Shell:** Bash
 - **User:** `analyst`
-

Task 1: Generate Output with `echo`

The first task was to use the `echo` command to display text in the terminal.

I started with:

```
echo hello
```

The output was:

```
hello
```

This helped me understand that `echo` can be used to display text in the Bash terminal.

I also tried:

```
echo "hello"
```

The output was again:

```
hello
```

In this simple example, using quotation marks did not change the output.

I then used the command to display a name:

```
echo "Your name"
```

The command displayed the name as output in the terminal.

Task 2: Generate Output with `expr`

The second task was to use the `expr` command for basic calculations.

The first example was related to security alerts.

There were **32 alerts**, but only **8 alerts required action**. I used the following command:

```
expr 32 - 8
```

The result was:

```
24
```

This means that 24 alerts did not require action.

I then used another calculation related to login attempts.

The example used **3,500 login attempts per month**. To find the expected number of login attempts in one year, I used:

```
expr 3500 \* 12
```

The result was:

```
42000
```

So, 3,500 login attempts per month would result in approximately **42,000 login attempts in one year**.

I learned that `expr` can be used for basic integer calculations in the Bash shell.

Task 3: Clear the Terminal

The third task was to use the `clear` command.

I entered:

```
clear
```

This cleared the commands and output that were currently visible in the terminal.

The command is useful when I want to continue working with a clean terminal screen.

Additional Practice

The activity also allowed me to practice the commands further.

For example, I used:

```
echo "Example text"
```

which displayed:

```
Example text
```

I also learned that `expr` can perform basic mathematical operations such as:

```
+    Addition
-    Subtraction
*    Multiplication
/    Division
```

For example:

```
expr 20 + 10
```

returns:

```
30
```

One important thing I noticed is that spaces are needed between the numbers and operators when using `expr`.

Commands Practiced

Command	Purpose
<code>echo</code>	Display text in the terminal
<code>expr</code>	Perform basic calculations
<code>clear</code>	Clear the terminal screen

What I Learned

From this activity, I learned some basic Linux Bash commands and how commands work through the terminal.

I practiced displaying text using `echo`, performing calculations using `expr`, and clearing the terminal using `clear`.

Although these are basic commands, they helped me become more familiar with the Linux command line.

Cybersecurity Connection

Linux and the command line are important parts of cybersecurity.

Security professionals often use Linux terminals for different tasks such as checking logs, analyzing systems, troubleshooting problems, running security tools, and investigating security incidents.

This activity gave me a basic starting point for working with the Linux command line. I will build on these skills in future cybersecurity activities and labs.

Demonstration

The following screenshots demonstrate my practical work completed during this activity. They show the Bash terminal, the commands I practiced, and the output generated by the commands.

```
analyst@a12f20ff8d77:~$ echo Hello
Hello
analyst@a12f20ff8d77:~$ echo "Hello"
Hello
analyst@a12f20ff8d77:~$ echo "Abdu Samad Shah"
Abdu Samad Shah
analyst@a12f20ff8d77:~$ expr 32 - 8
24
analyst@a12f20ff8d77:~$ expr 3500 * 12
42000
analyst@a12f20ff8d77:~$ clear
```

```
analyst@a12f20ff8d77:~$ echo "Good Bye"
Good Bye
analyst@a12f20ff8d77:~$
```

The screenshots demonstrate my hands-on practice with:

- `echo`
 - `expr`
 - `clear`
 - Basic Bash input and output
 - Basic mathematical calculations
-

Conclusion

This activity gave me practical experience with the Linux Bash shell.

I practiced the `echo`, `expr`, and `clear` commands and learned how to enter commands and understand the output returned by the terminal.

These basic Linux skills will be useful as I continue learning cybersecurity and move toward more practical Linux, networking, and security-related activities.

Course Information

Google Cybersecurity Certificate

Course: Tools of the Trade: Linux and SQL

Activity: Working with the Bash Shell

Disclaimer

This document is part of my personal cybersecurity learning portfolio. The activity was completed for educational purposes as part of the Google Cybersecurity Certificate.